table/curve is provided in the Guidelines 3 for each ship
type defined in paragraph 1, and expressed as a function
of Capacity (e.g. deadweight). The "Standard fw "
table/curve is based on data of actual speed reduction of
as many existing ships as possible under the
representative sea condition.

fw and attained EEDI weather, if calculated, with the representative sea

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SOURCE: MEPC.212(63).pdf
table/curve is based on data of actual speed reduction of
as many existing ships as possible under the
representative sea condition.

fw and attained EEDI weather, if calculated, with the representative sea
conditions under which those values are determined, should be indicated in
the EEDI Technical File to make a distinction with the attained EEDI
calculated under regulations 20 and 21 of MARPOL Annex VI.

.10 feff(i) is the availability factor of each innovative energy efficiency technology.
feff(i) for waste energy recovery system should be one (1.0)4.

.11 fi is the capacity factor for any tech nical/regulatory limitation on capacity,
and should be assumed to be one (1.0) if no necessity of the factor is
granted.

.1 The capacity correction factor, fi, for ice-classed ships should be
taken as the lesser value of fi0 and fi,max as tabulated in Table 2, but
not less than fi,min = 1.0. For further information on approximate
correspondence between ice classes, see HELCOM
Recommendation 25/75.

Table 2: Capacity correction factor fi for ice-classed ships
Ship type fi0
fi,max depending on the ice class
IA Super IA IB IC
Tanker capacityLPP
331 . 3

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SOURCE: MEPC.212(63).pdf
correspondence between ice classes, see HELCOM
Recommendation 25/75.

Table 2: Capacity correction factor fi for ice-classed ships
Ship type fi0
fi,max depending on the ice class
IA Super IA IB IC
Tanker capacityLPP
331 . 3
00138 . 0 11 . 010 . 2PPL 08 . 071 . 1PPL 06 . 047 . 1PPL 04 . 027 . 1PPL

Bulk carrier capacityLPP
123 . 3
00403 . 0 11 . 010 . 2PPL 09 . 080 . 1PPL 07 . 054 . 1PPL 05 . 031 . 1PPL
General
cargo ship capacity
LPP 625 . 20377 . 0 11 . 018 . 2PPL 08 . 077 . 1PPL 06 . 051 . 1PPL 04 . 028 . 1PPL
Containership capacityLPP 329 . 21033 . 0 11 . 010 . 2PPL 08 . 071 . 1PPL 06 . 047 . 1PPL 04 . 027 . 1PPL
Gas carrier capacityLPP 590 . 20474 . 0 25 . 1 12 . 010 . 2PPL 08 . 060 . 1PPL 04 . 025 . 1PPL
Note: containership capacity is defined as 70% of the DWT.

3 Guidelines for the calculation of the coefficient fw for the decrease of ship speed in respective sea conditions will be developed.
4 EEDI calculation should be based on the normal sea-going condition outside Emission Control Area designated under paragraph 6 of regulation 13 in MARPOL ANNEX VI.

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SOURCE: MEPC.212(63).pdf
conditions will be developed.
4 EEDI calculation should be based on the normal sea-going condition outside Emission Control Area designated under paragraph 6 of regulation 13 in MARPOL ANNEX VI.
5 HELCOM Recommendation 25/7 may be found at http://www.helcom.fi .
RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
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.2 f i VSE for ship specific voluntary structural enhancement is expressed by the following formula:

designenhanced
designreference
VSEi DWT
DWTf
 Where:

designreferenceshipdesignreference t lightweighDWT

designenhancedshipdesignenhanced t lightweighDWT

For this calculation the same displacement ( ) for reference and enhanced design should be taken.

DWT before enhancements (DWTreference design) is the deadweight prior to application of the structural enhancements. DWT after enhancements (DWTenhanced design) is the deadweight following the application of voluntary structural enhancement. A change of material (e.g. from aluminum alloy to steel) between reference

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SOURCE: MEPC.212(63).pdf
enhancements (DWTenhanced design) is the deadweight following the application of voluntary structural enhancement. A change of material (e.g. from aluminum alloy to steel) between reference design and enhanced design should not be allowed for the fi VSE calculation. A change in grade of the same material (e.g. in steel type, grades, properties and condition) should also not be allowed.

In each case, two sets of structural plans of the ship should be submitted to the verifier for assessment. One set for the ship without voluntary structural enhancement; the other set for the same ship with voluntary structural enhancement. (Alternatively, one set of structural plans of the reference design with annotations of voluntary structural enhancement should also be acceptable.) Both sets of structural plans should comply with the applicable regulations for the ship type and intended trade.

.3 for bulk carriers and oil tankers, built in accordance with Common Structural Rules (CSR) of the classification societies and assigned the class notation CSR, the following capacity correction factor fiCSR should apply:

fiCSR = 1 + (0.08 · LWTCSR / DWTCSR)

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SOURCE: MEPC.212(63).pdf
Structural Rules (CSR) of the classification societies and assigned the class notation CSR, the following capacity correction factor fiCSR should apply:

fiCSR = 1 + (0.08 · LWTCSR / DWTCSR)
Where, DWTCSR is the deadweight determined by paragraph 2.4 and LWTCSR is the light weight of the ship.

.4 for other ship types, fi should be taken as 1.0.

.12 fc is the cubic capacity correction factor and should be assumed to be one (1.0) if no necessity of the factor is granted.

.1 for chemical tankers, as defined in regulation 1.16.1 of MARPOL Annex II, the following cubic capacity correction factor fc should apply:

fc = R 0.7 0.014, where R is less than 0.98
or
fc = 1.000, where R is 0.98 and above;
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Adopted on 2 March 2012
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where: R is the capacity ratio of the deadweight of the ship (tonnes) as determined by paragraph 2.4 divided by the total cubic capacity of the cargo tanks of the ship (m3).

.2 for gas carriers having direct diesel driven propulsion system

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SOURCE: MEPC.212(63).pdf
where: R is the capacity ratio of the deadweight of the ship (tonnes) as determined by paragraph 2.4 divided by the total cubic capacity of the cargo tanks of the ship (m3).

.2 for gas carriers having direct diesel driven propulsion system constructed or adapted and used for the carriage in bulk of liquefied natural gas, the following cubic capacity correction factor fcLNG should apply:

fcLNG = R -0.56